

Nabla v5

A regime-aware factor book with a rule-based cash dial
v4 with the red-team changes applied, scoped to fit the time left

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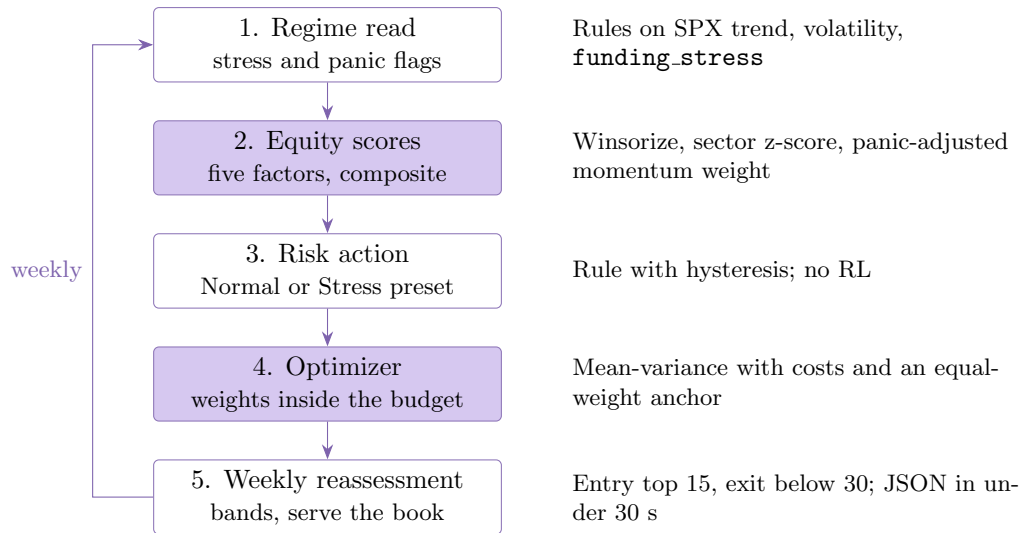
Status. Design document. Nothing has been run on the dataset yet; every performance statement is a hypothesis to test. v5 keeps v4's factors, regime-conditional momentum, optimizer with an equal-weight anchor, entry and exit bands, and research basis. It drops the RL bandit, the put overlay, partial trading, the HMM and the logistic calibration, which changed the book little and cost about five hours.

1 Changes from v4

v4 component	v5	Reason
RL bandit over six actions	Cut	The stress guardrail already forces A4–A6 when the rule fires, so outside stress the bandit only chooses among A1–A3, which differ by 0.05 of beta and 0–5% cash. With full feedback it is six ridge regressions and an argmax, not RL. The counterfactual reward table ignores path dependence (puts held 35–60 days, costs depend on last week's action), and a double-loss reward biases it defensive under a return-first score.
Put overlay	Cut unless SPY options exist	SPY is probably not in the universe. Puts on the three largest holdings sized to $h\beta_p E$ put about \$490k of notional on \$250–300k of stock at A4: a net bearish bet, not protection. At 35% implied volatility a 6% out-of-the-money 45-day put costs about 2.3% of notional, roughly 1.1% of the book in one purchase.
Partial trading, $\tau = 0.5$	Cut	The optimizer already charges trading costs, so τ double-counts them, and a new top pick would get only half its weight in week 1 of a 4-week test.
Two-state HMM	Replaced by rules	Its only remaining uses were the bandit context and the panic flag. A rule does both with fewer moving parts.
Logistic calibration $p_{i,t}$	Chart only	It does not change the ranking, and the optimizer uses the score, not p . Kept as a pitch chart if time allows.
Candidate pool = top 25	Top 25 $\cup$ held names ranked ≤ 30	Otherwise a held name ranked 26–30 cannot stay, contradicting the exit band.
50/50 blend after solving	Blend inside the optimizer	Blending after the solve can break sector, beta and ADV limits.
Trailing stop	Optional	Fires only for names ranked 21–30 that are also down 15%; rare in four weeks.

2 What the model does

Before the test window opens, and again every week inside it, the model runs five steps on the newest data. Steps 2 and 4 use fitted parameters; nothing is refit during the test.



3 What the research says

Topic and source	Finding	What v5 does
Volatility timing (Moreira and Muir, 2017; Cederburg et al., 2020)	Cutting risk in high volatility raised Sharpe in sample; real-time versions underperformed for most of 103 strategies, though momentum and profitability held up better.	A coarse two-level cash dial, thresholds fixed in advance; no fine volatility scaler.
Momentum crashes (Daniel and Moskowitz, 2016)	Momentum crashes cluster in rebounds after declines with high volatility.	Halves the momentum weight in the panic state, decided before testing.
Short-term reversal (Jegadeesh, 1990; Lehmann, 1990)	Last week's and last month's losers tend to beat recent winners over the next week or month.	Never sells a name only because it fell. Momentum skips the last month.
Stop-losses (Kaminski and Lo, 2014)	Under a random walk, stop-losses lower expected return; they help only when returns trend.	Exits are driven by score decay, not price alone.
Protective puts (Israelov, 2019)	Cutting equity exposure beat buying puts on drawdown per unit of return, except when a crash lands before expiry.	Cash is the hedge. No puts unless index options exist.
Estimation error (DeMiguel, Garlappi and Uppal, 2009)	None of 14 optimizers consistently beat equal weighting out of sample.	Heavy shrinkage and an equal-weight anchor inside the optimizer; the optimizer must beat equal weight to ship.
Trading costs (Gârleanu and Pedersen, 2013)	With costs, the best policy trades toward an aim portfolio rather than jumping to it.	Cost term in the objective plus entry and exit bands.
Deep RL portfolios (Velay et al., 2023)	Most deep RL portfolio agents generalized poorly and degraded in backtests.	No RL; the rule is the controller.

4 The five steps in detail

4.1 Step 1: regime read

All inputs use data on or before the decision date.

- **Stress** is on when SPX is below its 200-day average *and* either 21-day SPX realized volatility or `funding_stress` is above its 80th percentile (expanding history).
- **Panic** is on when the SPX 12-month return is negative *and* 21-day realized volatility is above its 80th percentile. It only changes the momentum weight in Step 2.
- **Hysteresis**. A flag turns off only after its conditions have been false for two straight weeks.

4.2 Step 2: equity scores

After the liquidity filter (20-day dollar volume above \$50M, price above \$5, the most illiquid Amihud decile removed), each stock gets the five v3 factors: 12-1 momentum, guidance velocity, quality, value (earnings yield) and the volatility premium $-\log(\text{IV}_{30}/\text{RV}_{21})$. Each factor is winsorized at the 1st and 99th percentiles and z-scored within about 10 sector groups.

$$s_{i,t} = \frac{\sum_{k \in K_{i,t}} a_k(t) z_{i,t}^{(k)}}{\sum_{k \in K_{i,t}} a_k(t)}, \quad a_{\text{mom}}(t) = \begin{cases} 0.5 & \text{panic} \\ 1 & \text{otherwise} \end{cases}, \quad a_k = 1 \text{ for the other factors.}$$

Candidate pool. The top 25 names by $s_{i,t}$, plus any currently held name still ranked 30 or better, with at most 3 names per correlation cluster. Apply the crowding check against a naive 12-month momentum top 20 and swap marginal overlaps for the next-best names.

4.3 Step 3: risk action

Preset	When	Cash floor	Beta band	Notes
Normal	Stress off	0–5%	1.10–1.20	Default
Stress	Stress on	25%	0.90–1.00	Highest-beta names swapped for lower-beta ones

If the judges keep the 1 ± 0.01 weight-sum check, the Stress preset holds low-beta names instead of cash.

4.4 Step 4: optimizer

Expected returns with heavy shrinkage:

$$\mu_i = \tilde{\beta}_i m + \lambda_\alpha \hat{\rho} \sigma_{cs} \hat{s}_i, \quad \tilde{\beta}_i = 0.67 \hat{\beta}_i + 0.33,$$

where $m \approx 0.8\%$ per 21 days, $\hat{\rho}$ is the out-of-sample rank IC, σ_{cs} the cross-sectional standard deviation of 21-day returns, $\hat{s}_i$ the standardized score and $\lambda_\alpha \in (0, 1]$ a shrinkage factor. Covariance $\hat{\Sigma}$ uses Ledoit-Wolf shrinkage on 252 daily returns.

Problem (cvxpy, over the candidate pool):

$$\max_w \mu^\top w - \frac{\gamma}{2} w^\top \hat{\Sigma} w - \sum_i c_i |w_i - w_i^{\text{prev}}| - \kappa \|w - w^{\text{EW}}\|_2^2$$

subject to

- $\mathbf{1}^\top w \leq 1 - c_{\text{floor}}$, with the cash floor from the preset;
- $0 \leq w_i \leq 0.10$, and each sector group at most 0.30;
- $\beta_L \mathbf{1}^\top w \leq \tilde{\beta}^\top w \leq \beta_H \mathbf{1}^\top w$ (the preset's beta band);
- $w_i V \leq 0.01 \text{ADV}_i$, and each name reporting earnings in the window at most 0.06.

The anchor term $\kappa \|w - w^{\text{EW}}\|^2$ pulls the solution toward equal weights over the top 15 at the same invested total; it replaces v4's post-solve 50/50 blend, so every constraint holds in the final weights. Set κ so the solution sits roughly halfway between the unanchored optimum and equal weight, and γ so predicted volatility is about 18–22% a year.

Name count. Solve, drop weights below 2%, re-solve on the survivors; keep 12 to 18 names.

Fallback. If the solver fails or is infeasible at request time, serve the equal-weight top 15 with caps. Wrap the solve in a timeout.

4.5 Step 5: weekly reassessment

- Rescore every stock and recompute the flags and preset.
- **Entry band:** a new name enters only if it ranks in the top 15.
- **Exit band:** a held name leaves only if it falls below rank 30.
- **Trades:** go directly to the optimizer's target; skip any change under 2 percentage points. The cost term already discourages small moves.
- **Winners:** no profit-taking while the score holds; trim back to 10% only if drift pushes a name above 12%.
- **Earnings:** keep the 6% cap through a report and do not add in the week before it.
- **Optional trailing stop:** exit if a name is 15% below its high since entry and ranked worse than 20.
- Set the as-of date to `max(ds.trading_days())`, never `date.today()`. If no new data arrives, the book is unchanged.

5 Costs and validation

Costs. Every trade pays $c_i = |\Delta w_i| V \left(\frac{1}{2} s_i + k A_i |\Delta w_i| V \right)$, where s_i is 5, 10 or 20 bps by liquidity tercile, A_i is 21-day Amihud illiquidity and $k = 1$. Rerun everything at twice these costs.

Backtest. The exact weekly loop from 2018 to the cutoff. Anything fitted uses only earlier data, labels lag 21 trading days, and the last six months stay untouched until the go/no-go. One return function matches `check.py` and is reused by `/backtest`.

Metrics against SPX and an equal-weight liquid universe: total and annualized return, Sharpe, maximum drawdown, turnover and cost paid; rolling 21-day returns (median, 5th and 95th percentiles, worst window); sub-periods 2018 Q4, 2020 and 2022.

Ablation ladder. Each rung must beat the one before it on walk-forward folds after costs.

1. v3 equal-weight top 15 with limits and bands
2. add the panic momentum weight
3. add the rule-based cash dial
4. add the optimizer with the equal-weight anchor

Leak tests. Shuffled labels should erase the edge; delaying signals one day should hurt; dropping one factor at a time should show each factor earns its place.

Survivorship. The universe is today's survivors, which flatters concentrated high-beta books. State it in `AUDIT.md`; never present backtest returns as expected returns.

6 Pitfalls and solutions

Pitfall	Why it hurts	Solution
Regime lag, few crises	Regimes are recognized late; only a handful of stress episodes exist.	Coarse presets, thresholds fixed before testing, hysteresis.
Momentum crash in rebounds	Momentum loses badly when the market snaps back after a high-volatility decline.	Halve momentum weight in panic; quality and value diversify.
Crowding	Books held by many teams move together.	Crowding check against a naive momentum top 20.
Estimation error	Optimizers amplify noise; equal weight is hard to beat.	Shrunk μ , Ledoit-Wolf, caps, equal-weight anchor; ship rung 3 if rung 4 fails.
Solver failure live	An infeasible problem breaks the endpoint.	Offline feasibility check, timeout, equal-weight fallback.
Cutting losers into a bounce	Short-term reversal.	Exits on score decay, not price.
Churn and cost drag	Weekly trading on noisy scores.	Bands, 2-point no-trade band, cost term.
Leaks	Unfiltered fundamentals flatter backtests.	Filing-date gating and the leak tests.
Weight-sum check	Public <code>check.py</code> still requires 1 ± 0.01 .	Ask; if unchanged, Stress holds low-beta names, not cash.

7 Timeline

Clock times start at 6:45 PM Saturday and assume submission around noon Sunday. Each gate is a deployable submission, so a cut never leaves us empty-handed.

Time	Ezekiel (quant)	Jesse (finance)	Gate
6:45–7:15 PM	Repo and fixed interfaces (<code>config/model.json</code> , weights JSON); token check	Send organizer questions; check "SPY" in <code>ds.universe()</code>	Interfaces agreed; puts in or out
7:15–10:00	Loader anchored to last data date; split and dividend adjustment; deploy starter; <code>check.py</code>	Sector map, liquidity filter, quality and value factor code (with Claude), cost function	G1: public URL scores 100; splits clean
10:00 PM–1:30 AM	Weekly backtest with costs; composite; top 15 with limits and bands; wire live weekly scoring with this v1	Factor signs and IC; holdings review; crowding check; <code>AUDIT.md</code> outline	G2: v1 live; beats equal weight after costs
1:30–4:00 AM	Panic weight and cash rule (rungs 2–3); optimizer with anchor (rung 4)	Charts script (equity curve, drawdown, sector exposure); ablation table	G3: best passing rung deployed
4:00–7:30 AM	Sleep	Sleep	
7:30–9:30 AM	Go/no-go on held-back six months; freeze <code>model.json</code> ; redeploy; rerun <code>check.py</code>	Finish <code>AUDIT.md</code> and <code>README</code>	G4: frozen; endpoint under 30 s
9:30–11:30 AM	Buffer; fix anything found	Pitch slides; rehearse twice	
11:30 AM–noon	Submit	Submit	Done

Meals at about 9 PM and 1:30 AM, inside whatever block is running.

Cut rules (decided now)

1. Guidance velocity gets 45 minutes. If the timestamps do not parse, drop to four factors.
2. If G2 slips past 2 AM, skip the optimizer and ship v1 with the cash rule.
3. If rung 4 does not beat rung 3 by 3:30 AM, ship rung 3.
4. Nothing new starts after 4 AM.
5. Puts only if SPY and its options are in the universe *and* an option leg passes `check.py`. Otherwise cash only.

Where time usually leaks

Point-in-time fundamentals and data loading; deploy surprises (sleeping free tiers, token setup); and live weekly scoring, which is why it is wired at G2 with the simple model rather than at the end.

8 Open questions for the organizers

1. How does new weekly data reach the endpoint during the test?
2. Does the holdings check now accept weights summing below 0.99?
3. Besides total return, which metrics count and with what weight?
4. Are SPY or any index ETF and its options in the tradable universe?

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